

Installation Instructions for Honeywell Experimental Products

ISSUE 3
PK 88800

NOTICE

These products are prototype, preproduction items that have yet to complete all phases of product release testing. Designated by the letter "X" within their part number identification, these items are sold **"AS IS" WITH NO WARRANTY, EXPRESS OR IMPLIED, INCLUDING THAT OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.**

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PK88800-3-EN IL50 GLO
January 2011
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Honeywell

INSTALLATION INSTRUCTIONS FOR THE **BATTERY SAFETY ELECTROLYTE VAPOR SENSOR** **X123932BES PROOF-OF-CONCEPT PROTOTYPE**

3013-9781
Issue 3

1.0 GENERAL INFORMATION

The Honeywell BES Series Battery Safety Electrolyte Vapor Sensor is an automotive-grade battery safety sensor intended for use with lithium-ion batteries. The sensor is designed to detect the presence of electrolyte solvent vapors that are typically released during first vent, and have been observed to occur 5 to 15 minutes prior to thermal propagation in the form of smoke or fire.

2.0 USE OF PROOF-OF-CONCEPT PROTOTYPE

The intended use of the Honeywell BES Series Proof-of-Concept prototype is to demonstrate the sensing performance of the planned production series sensor. The Proof-of-Concept prototype is not intended for on-vehicle test. Honeywell recommends that the Proof-of-Concept prototype be subjected to lab-bench type testing only.

3.0 INSTALLATION

Mount the sensor using the two mounting bushings and two M5 bolts. The mounting tabs accommodate various bolt head styles, such as hex, socket head cap, and hex flange bolt head. Torque mounting bolts to 5 Nm to 7 Nm.

The sensor incorporates a Molex® 349129040 connector that meets US CAR 050-U-004-1-A02 specifications. For mating connector, use Molex® P/N 347910040.

4.0 SPECIFICATIONS

See Table 1.

BATTERY SAFETY ELECTROLYTE VAPOR SENSOR X123932BES PROOF OF CONCEPT PROTOTYPE

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TABLE 1. SPECIFICATIONS

Characteristic	Parameter
General	
Detection principle	Metal Oxide
Startup time	70 seconds to valid output
Response time	3 seconds (typical)
Sensed media (Typical battery electrolytes)	EMC (Ethyl Methyl Carbonate) DEC (Diethyl Carbonate) DMC (Dimethyl Carbonate) EC (Ethylene Carbonate)
Environment	
Operating temperature range	-40°C to 85°C [-40°F to 185°F]
Storage temperature range	-40°C to 85°C [-40°F to 185°F]
Humidity	0 %RH to 90 %RH
Operating pressure	-100 m to +4,600 m altitude
Shock	50 g (3 drops in each +/- direction, 3 axes)
Sealing	NEMA IP3X
Electrical	
Supply voltage	8 Vdc to 16 Vdc
Current consumption	<25 mA average
Communication	CAN 2.0B/J1939
Baud rate	250 kbps

TABLE 2. INSTALLATION SPECIFICATIONS

Characteristic	Parameter
Connector type	Molex® 349129040/US CAR 050-U-004-1-A02 Mating connector: Molex® 347910040
Weight	30 g max.
Mounting bolt	M5, installation torque 5 Nm to 7 Nm

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Figure 1. Body Dimensions (for reference only), mm

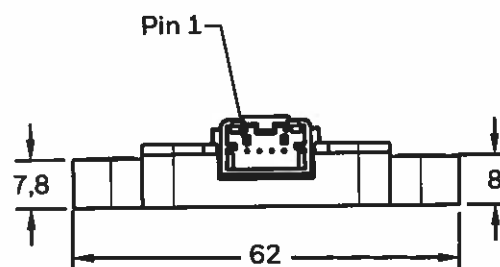
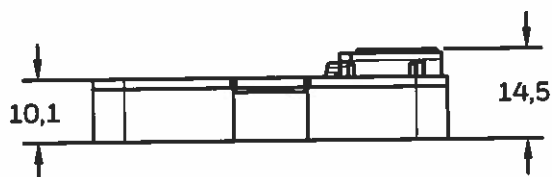
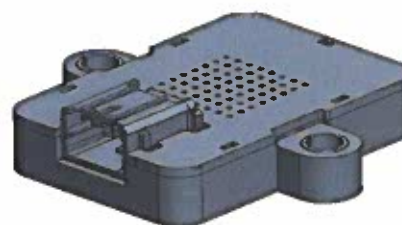
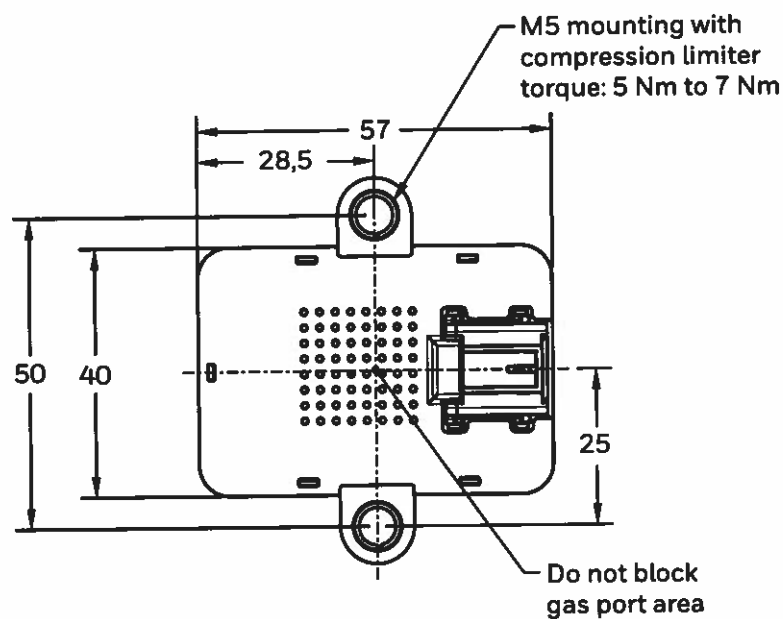


TABLE 3. PINOUT

Pin	Function
Pin 1	Power in
Pin 2	Ground
Pin 3	CANH
Pin 4	CANL

NOTICE

Product is a non-sealed device. If the sensor is used in a condensing environment, the sensor should be oriented to prevent moisture from collecting in the sensor gas port area.

5.0 CAN SENSOR COMMUNICATION

5.1 BES STATE FORMAT

TABLE 4. PDU 0XFF01/65281

Frame Format	29-Bit ID						Data (8 Bytes)							
							Scalar							
Field	P	EDP	DP	PF	PS	SA	State	Deg C	Low Byte	High Byte	Not Used			
Number of bits	2	1	1	8	8	8	8	8	8	8	8	8	8	8
CAN message	0x08			0xFF	0x01	0xE4	0x03	0x17	0x1C	0x00	0x00	0x00	0x00	0x00

State

The **State** byte represents the present state of the sensor:

0. Illegal
1. Error
2. Warmup
3. Normal
4. Alarm

Scalar Value

Value = (int16_t)Scalar/100 (-100.00 to 100.00)

Typical Values:

- -2.00 = typical warm up condition
- 0.00 = clean air
- 1.00 = alarm trigger level
- >4.00 = typical high level of electrolytes

Temperature

Temperature (°C) = (int8_t)DegC (-128°C to +127°C)

Broadcast Rate

This message is broadcast once every 1 second.

Example

Broadcast Normal Operation at 23°C (0x17h=23) and 0.28 Scalar (0x001C = 28)

ID	Data
8FF01E4x	03 17 1C 00 00 00 00 00

NOTICE

The X123932BES Battery Safety Electrolyte Vapor Sensor is a Proof-of-Concept prototype and should only be tested in a controlled engineering lab environment. The sensor incorporates an electronic volatile organic compound (VOC) gas sensing integrated circuit for detection of battery electrolyte vapors that are typically released during battery venting. The VOC gas sensing integrated circuit contains a metal oxide semiconductor element that is maintained at a temperature of 400°C whenever the sensor is powered. It is noted that the Proof-of-Concept prototype does not contain a flame arresting feature and that extreme caution should be used if the sensor is powered and/or subjected to testing in the presence of flammable levels of combustible gases.

WARRANTY/REMEDY

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⚠ WARNING PERSONAL INJURY

DO NOT USE these products as safety or emergency stop devices or in any other application where failure of the product could result in personal injury.

Failure to comply with these instructions could result in death or serious injury.

⚠ WARNING MISUSE OF DOCUMENTATION

- The information presented in this product sheet is for reference only. Do not use this document as a product installation guide.
- Complete installation, operation, and maintenance information is provided in the instructions supplied with each product.

Failure to comply with these instructions could result in death or serious injury.

For more information

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3013-9781-3-EN | 3 | 04/24
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